

Subject: Physics
Subject code: PH113
Modules: 5
Credit: 4
Maximum Marks: 100

Evaluation:
Test/Assignment: 10x5 = 50
End Sem Exam: = 50

(PH 113) PHYSICS

Module-1	Physical Optics: Polarization, Malus' Law, Brewster's Law, Double Refraction, Interference in thin films (Parallel films), Interference in wedge-shaped layers, Newton's rings, Fraunhofer diffraction by single slit, Double slit.	[9L]
Module-2	Electromagnetic Theory: Curl, Gradient, Divergence, Gauss theorem, Stokes theorem, Gauss's law, Applications, Concept of electric potential, Relationship between E and V, Polarization of dielectrics, dielectric constant, Boundary conditions for E & D, Gauss's law in magnetostatics, Ampere's circuital law, Boundary conditions for B & H, Equation of continuity of charge, Displacement current, Maxwell's equations.	[9L]
Module-3	Special Theory of Relativity: Introduction, Inertial frame of reference, Galilean transformations, Postulates, Lorentz transformations and its conclusions, Length contraction, time dilation, velocity addition, Mass change, Einstein's mass energy relation.	[9L]
Module-4	Quantum Mechanics: Planck's theory of black-body radiation, Compton effect, Wave particle duality, De Broglie waves, Davisson and Germer's experiment, Uncertainty principle, physical interpretation of wave function, Schrodinger equation in one dimension, free particle, particle in an infinite square well.	[9L]
Module-5	Lasers: Spontaneous and stimulated emission, Einstein's A and B coefficients, Population-inversion, Light amplification, Basic laser action, Ruby and He-Ne lasers, Properties and applications of laser radiation, Elementary ideas of fiber optics and application of fiber optic cables.	[9L]

Text books:

- T1:** A. Ghatak, Optics, 4th Edition, Tata Mcgraw Hill, 2009
T2: Mathew N.O. Sadiku, Elements of Electromagnetics, Oxford University Press, 2001
T3: Arthur Beiser, Concept of Modern Physics, 6th edition, Tata McGraw- Hill, 2009

Reference books:

- R1:** Fundamentals of Physics, Halliday, Walker and Resnick

Gaps in the syllabus (to meet Industry/Profession requirements) : NA

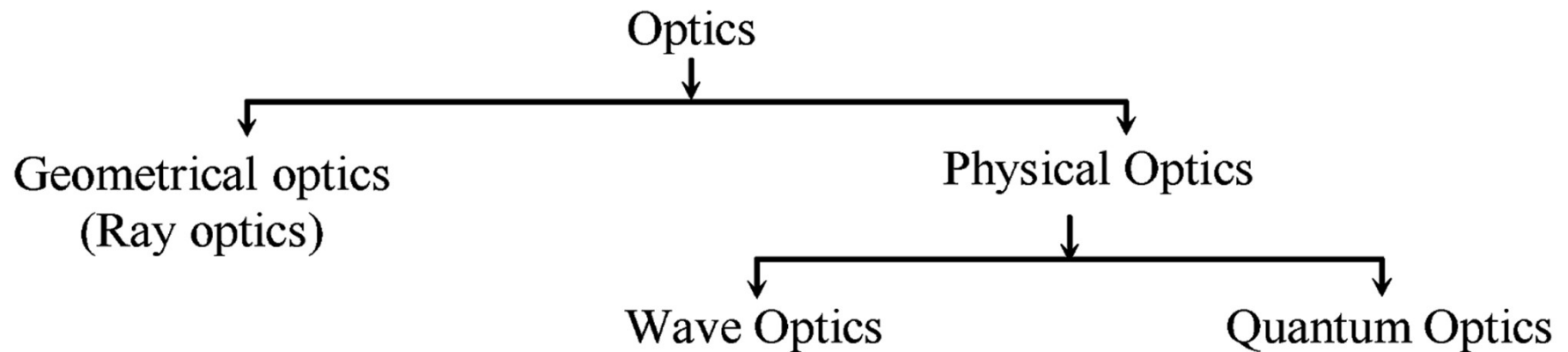
POs met through Gaps in the Syllabus : NA

Topics beyond syllabus/Advanced topics/Design : NA

PHYSICAL OPTICS

Dr. Anand Kumar Srivastava
Department of Physics
BIT Mesra, Jaipur Campus

Optics classification



1. **Ray Optics:** Straight line path, lenses, mirrors
(Reflection, Refraction, Dispersion of Light)
2. **Physical optics:** Light as wave: Interference, Diffraction, Polarization
3. **Quantum Optics:** Light as particle: Photo electric effect, Compton effect, Black body radiation

What is Light:

Light is a form of energy; its propagation is considered as transfer of energy from the luminous source (lamp, bulb) to the eye either directly or through any object.

To explain the energy transfer several theories have been proposed:

- (i) Newton's corpuscular (particle) theory
- (ii) Huygen's wave theory
- (iii) Maxwell's electromagnetic theory
- (iv) Quantum theory

Newton's Particle theory of light could not explain the experimentally observed phenomenon like interference, diffraction and polarization. These physical processes could be explained by **wave theory** (Huygen, Young, Fresnel, Fraunhofer).

Wave nature of light

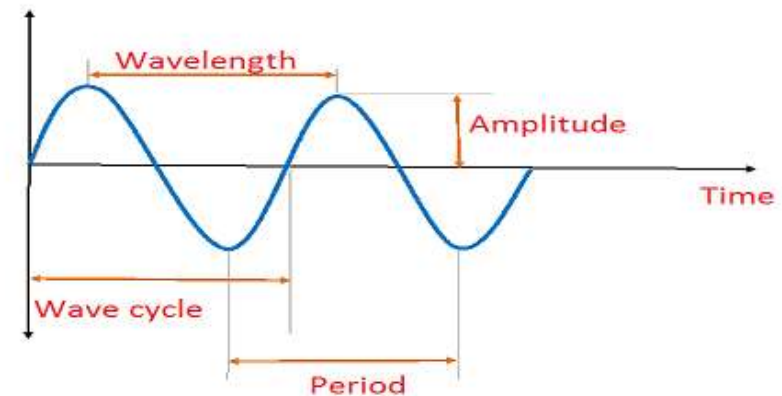
$$Y = A \sin (\omega t - Kx)$$

$\omega = 2\pi f = 2\pi / T$, angular frequency

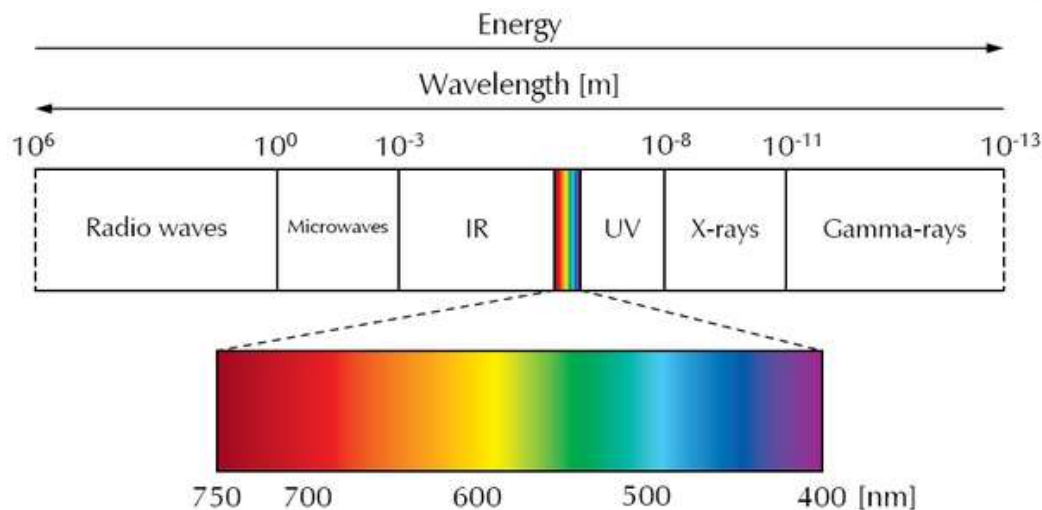
$K = 2\pi / \lambda$, propagation constant

$v = f \cdot \lambda$, wave velocity

$(\omega t - Kx)$, Phase of the wave



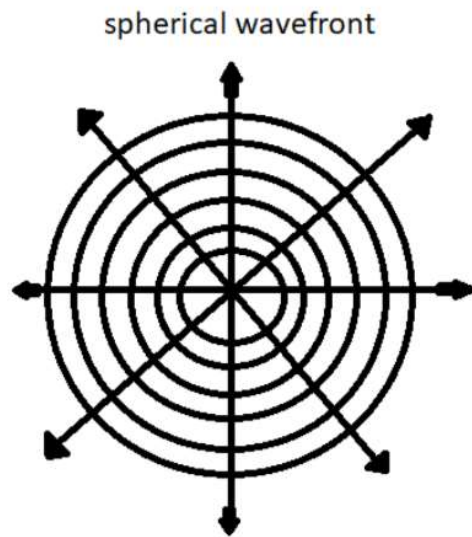
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Electromagnetic Spectrum

Wave Front

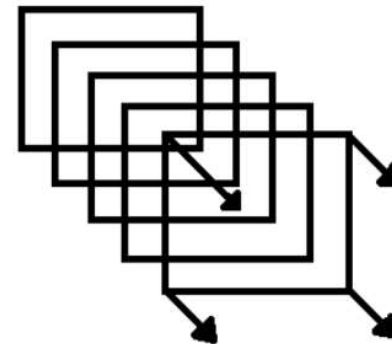
A wavefront is defined as the continuous locus of all the particles which are vibrating in the same phase.



cylindrical wavefront



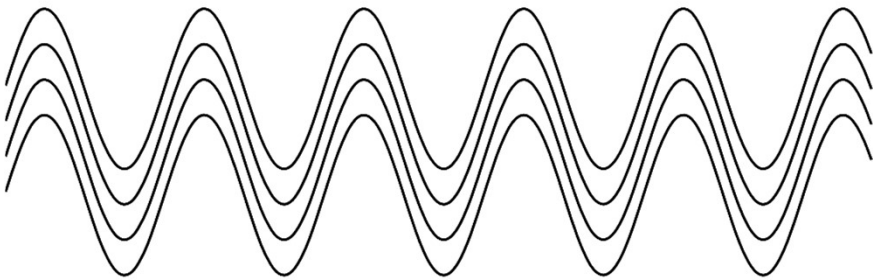
Plane wavefront



Coherent Source

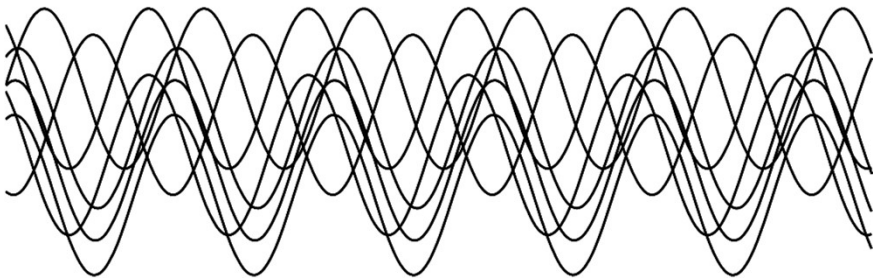
If the waves emitted from a light source produce same frequency wave and show constant phase difference between them, the light source is known as coherent source.

Coherent



LASER

Incoherent

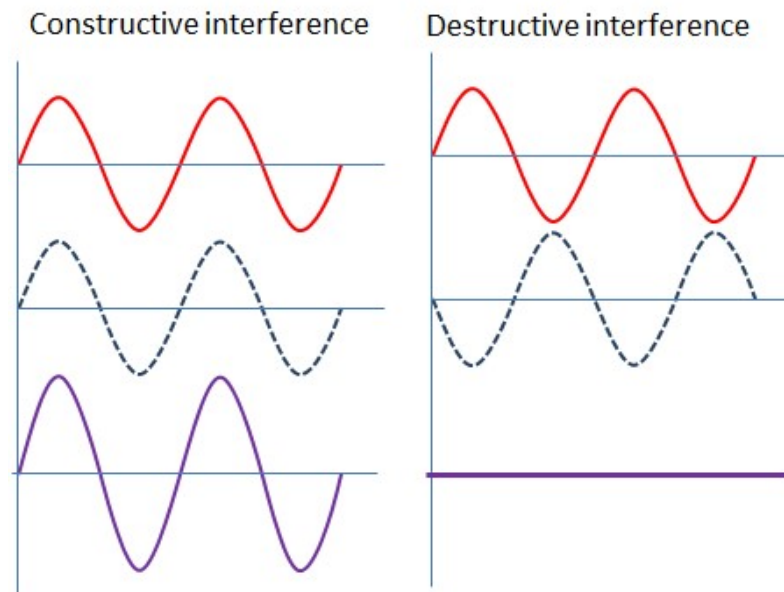


BULB

Interference of Light

When two waves of same frequency and having a constant phase difference superimpose on each other, then the resultant intensity of light is modified in the region of superposition. It is known as interference of light.

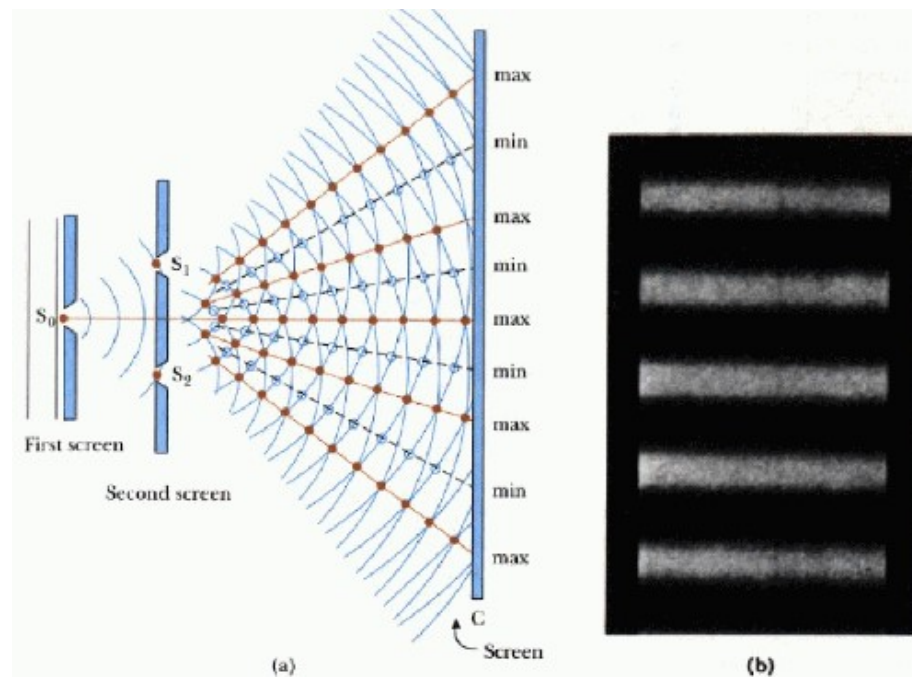
- Constructive interference
- Destructive interference



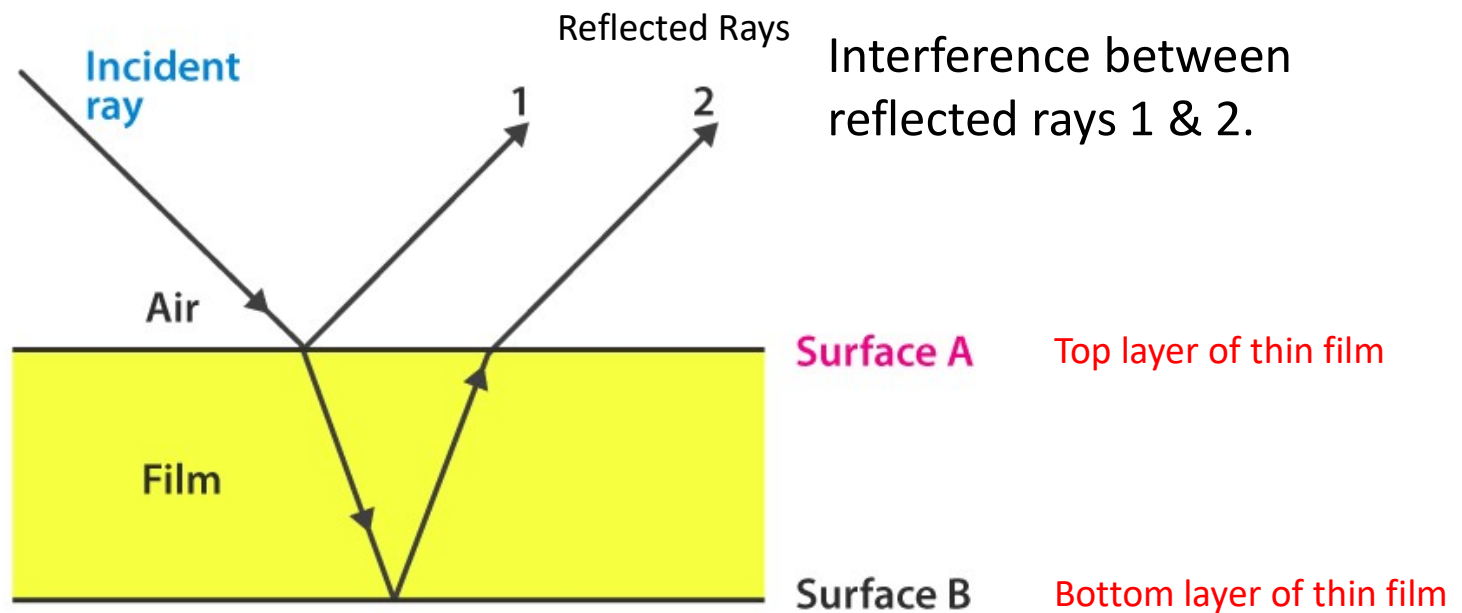
Methods to observe Interference of light

1. Interference by division of wavefront
2. Interference by division of amplitude

Young's double-slit
experiment:
Division of wavefront



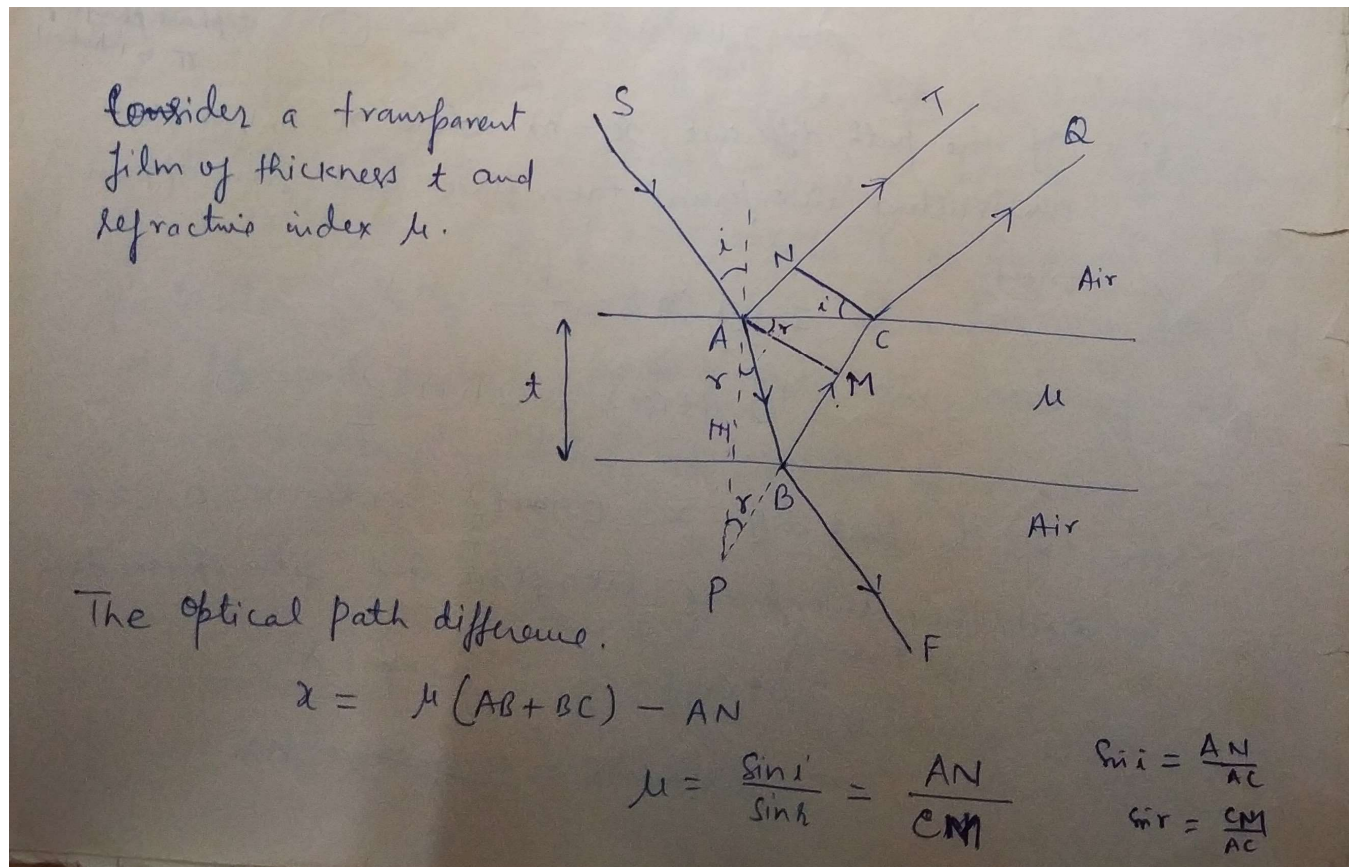
Thin film Interference: Division of amplitude (Intensity)



Colours are observed in soap bubbles due to thin film interference.



Interference due to parallel thin film using reflected light



$$AN = \mu \cdot CM$$

$$x = \mu(AB + BC) - \mu \cdot CM$$

$$= \mu(AB + BC - CM)$$

$$= \mu(PC - CM) = \mu \cdot PM \quad AB = BP$$

In $\triangle APM$,

$$\cos \theta = \frac{PM}{AP}$$

$$PM = AP \cdot \cos \theta = (AE + EP) \cos \theta$$

$$= 2 \cos \theta$$

$$\therefore x = \mu \times 2 \cos \theta$$

$$= 2\mu \cos \theta \quad \text{--- (1)}$$

The corrected path diff.

$$x = 2\mu \cos \theta - \frac{\lambda}{2}$$

(Electromagnetic th
a phase change
 π is introduced)
--- (2)

Conditions of maxima and minima

(i) If the path difference $x = n\lambda$ where $n = 0, 1, 2, 3, 4, \dots$, constructive interference takes place and film appears bright.

$$2\mu t \cos r - \frac{\lambda}{2} = n\lambda$$

$$2\mu t \cos r = (2n+1)\frac{\lambda}{2} \quad \text{--- (3)}$$

(ii) If the path diff. $x = (2n+1)\frac{\lambda}{2}$, where $n = 0, 1, 2, 3, \dots$, destructive interference takes place and film appears dark.

$$2\mu t \cos r - \frac{\lambda}{2} = (2n+1)\frac{\lambda}{2}$$

$$2\mu t \cos r = \cancel{(n+1)\lambda} n\lambda$$

Conditions of maxima and minima in thin film interference

Maxima (Bright Light)

$$2\mu t \cos r = (n + \frac{1}{2})\lambda, n = 0, 1, 2, 3, 4, 5, \dots$$

Minima (Dark Light)

$$2\mu t \cos r = n\lambda, n = 0, 1, 2, 3, 4, 5, \dots$$

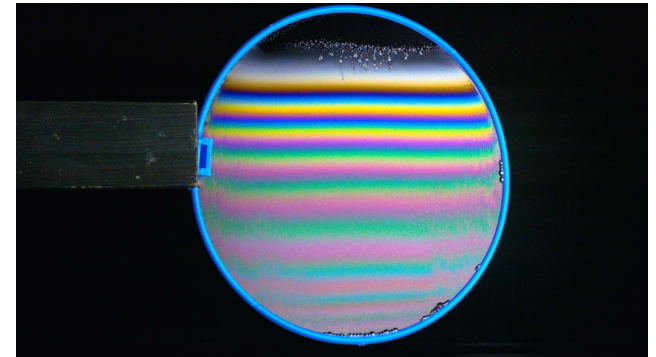
Where,

μ - Refractive index of thin film medium

t - thickness of thin film

λ - wavelength of light

r - angle of refraction



Interference due to wedge shaped thin film

For wedge shaped thin film,

Maxima Condition,

$$2\mu t \cos(r+\theta) = \left(n + \frac{1}{2}\right) \lambda$$

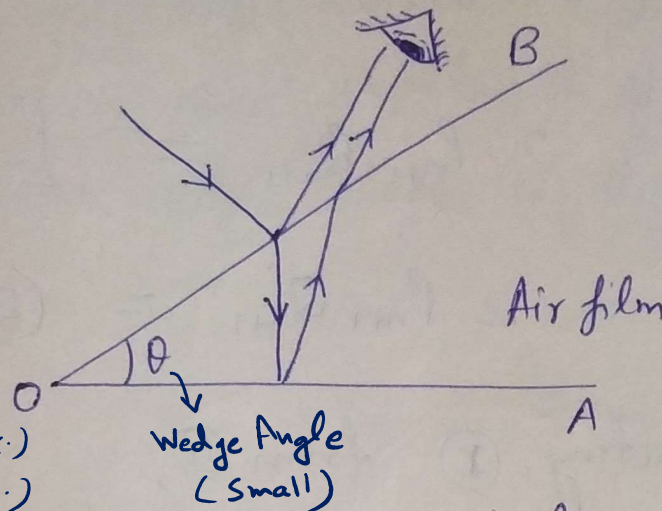
Minima Condition,

$$2\mu t \cos(r+\theta) = n\lambda$$

For small angle, $\theta \approx 0$

$$\text{Hence, } 2\mu t \cos r = \left(n + \frac{1}{2}\right) \lambda \quad (\text{Max.})$$

$$2\mu t \cos r = n\lambda \quad (\text{Min.})$$



Consider two planes OA and OB, inclined at angle θ and enclosing a wedge shaped air film. The thickness of the air film increases from O to A. When the air film is viewed with reflected monochromatic light, a system of equidistant interference fringes are observed which are parallel to the line of intersection of two surfaces.

Suppose the n th bright fringe occurs at P_n .
 The thickness of air film at P_n is $P_n Q_n$.

For small inclination angle,
 and near normal incidence
 $\cos r = 1$

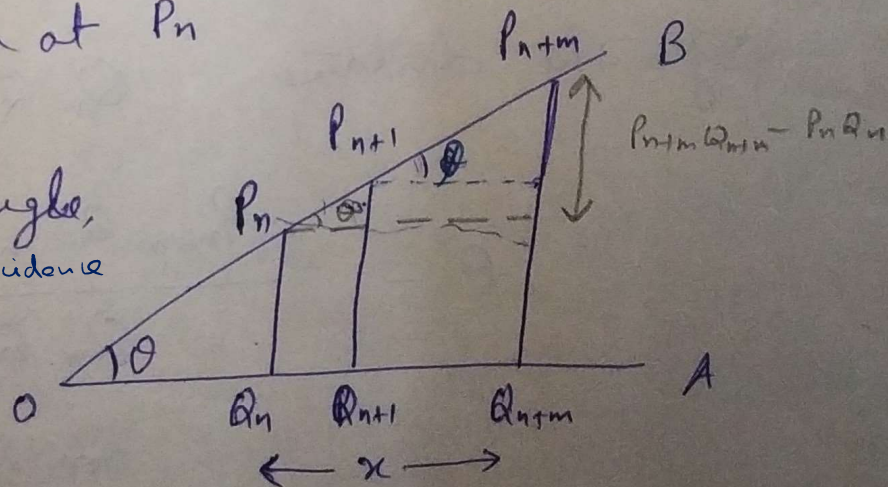
$\therefore$ Condition of Bright fringe,

$$2 \mu t \cos r = (2n+1) \frac{\lambda}{2} \quad n=0,1,2,3$$

Here, $\mu = 1$ (for air)

$$\cos r = 1$$

$$t = P_n Q_n$$



$$\therefore 2 P_n Q_n = (2n+1) \frac{\lambda}{2}$$

— (1)

The next bright fringe $(n+1)$ will occur at P_{n+1}

$$\therefore 2 P_{n+1} Q_{n+1} = [2(n+1)+1] \frac{\lambda}{2}$$

$$2 P_{n+1} Q_{n+1} = (2n+3) \frac{\lambda}{2} \quad \text{--- (2)}$$

Subtracting (1) from (2),

$$P_{n+1} Q_{n+1} - P_n Q_n = \frac{\lambda}{2} \quad \text{--- (3)}$$

Thus the next bright fringe will occur at the point where the thickness of air film increases by $\lambda/2$.

Suppose $(n+m)$ th bright fringe is at P_{n+m} .

$$\therefore P_{n+m} Q_{n+m} - P_n Q_n = \frac{m \cdot \lambda}{2}$$

If the distance,

$$Q_n Q_{n+m} = x$$

For small wedge angle,

$$\tan \theta \approx \theta,$$

$$\therefore \theta =$$

$$\frac{P_{n+m} Q_{n+m} - P_n Q_n}{Q_n Q_{n+m}}$$

$$= \frac{m \cdot \frac{\lambda}{2}}{x} = \frac{m \lambda}{2x}$$

or

$$x = \frac{m \lambda}{2 \theta}$$

fringe width,

$$\beta = \frac{x}{m} = \frac{\lambda}{2 \theta}$$

for air

For other medium,

$$\beta = \frac{\lambda}{2 \mu \theta}$$

$\mu \rightarrow$ ref. index
of medium

Q. Fringes of equal thickness are observed in a thin glass wedge of refractive index 1.52. The fringe spacing is 1 mm and wavelength of light is $5893 \text{ \AA}$. Calculate the wedge angle in seconds of an arc.

$$\mu = 1.52, \quad \lambda = 5893 \times 10^{-8} \text{ cm.}$$

$$\beta = 1 \text{ mm} = 0.1 \text{ cm.}$$

$$\beta = \frac{\lambda}{2\mu\theta}, \quad \theta = \frac{\lambda}{2\mu\beta} \quad (\text{radian})$$

$$= \left(\frac{5893 \times 10^{-8}}{2 \times 1.52 \times 0.1} \right) \text{ radian}$$

$$= \left(\frac{5893 \times 10^{-8}}{0.304} \right) \times \frac{60 \times 60 \times 180}{\pi} \text{ Sec. of an arc.}$$

$$= 39.96 \text{ seconds of an arc.}$$

$$\left[\begin{array}{l} \text{Angle units.} \\ 1 \text{ second of an arc} = \frac{\pi}{60 \times 60 \times 180} \text{ Radian} \end{array} \right]$$

$$1 \text{ minute of arc} = \frac{1}{60} \text{ degree}$$

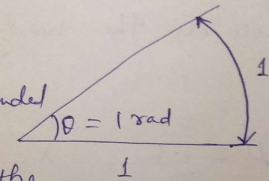
$$\therefore 3^\circ 30' = 3 + \frac{30}{60} = 3.5^\circ$$

$$1 \text{ second of arc} = \frac{1}{60} \text{ of a min of arc} = \left(\frac{1}{60} \times \frac{1}{60} \right)^\circ = \frac{1}{3600}^\circ$$

$$3^\circ 7' 30'' = 3 + \frac{7}{60} + \frac{30}{60 \times 60} = 3.125^\circ$$

Radian

Radian is the angle subtended by an arc of a circle that has the same length as the radius of circle.



$$1 \text{ full circle} = 2\pi \text{ radian}$$

$$1 \text{ radian} = \frac{180}{\pi} \text{ degrees}$$

$$\approx 57.2958 \text{ degrees}$$

Newton's Rings

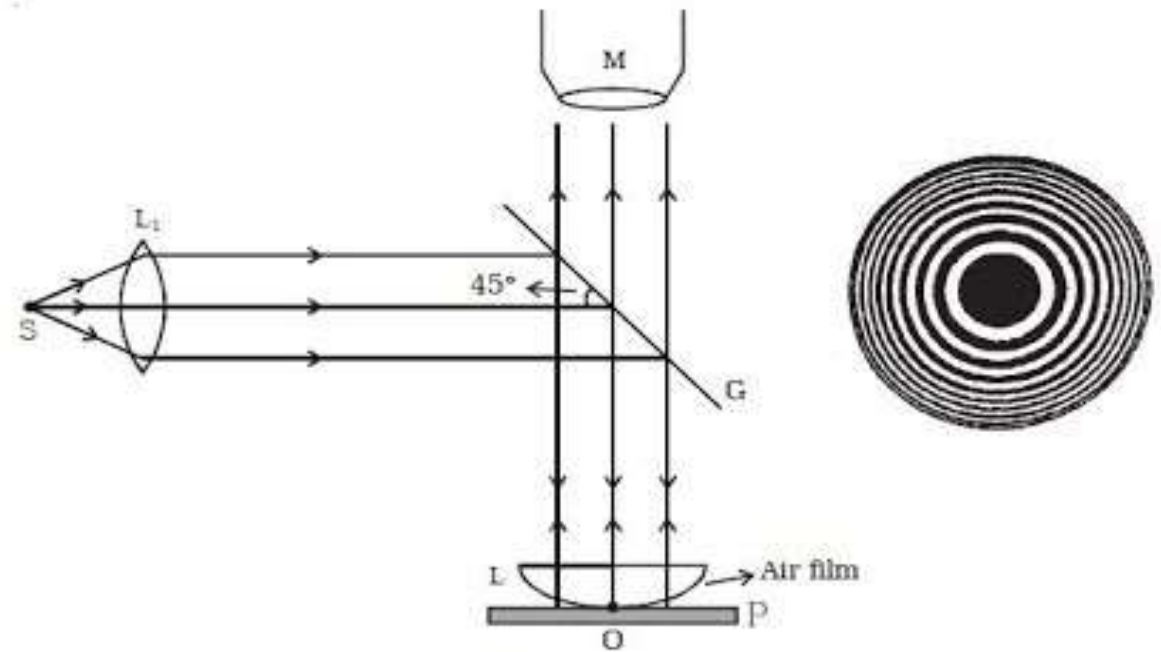
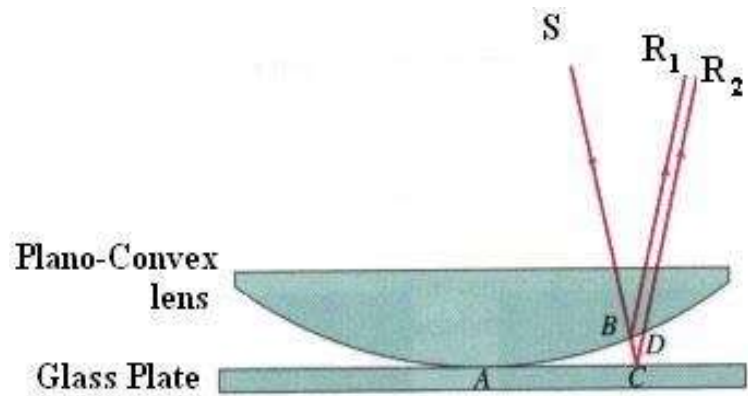
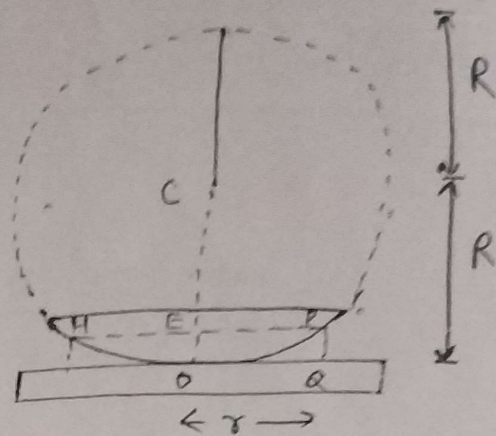
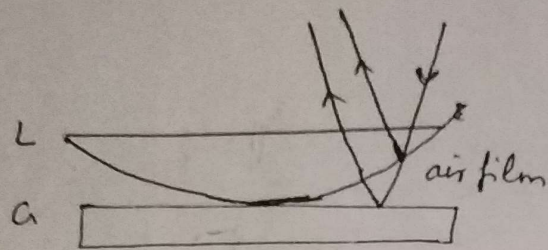


Fig 5.19 Newton's rings

Newton's rings by reflected light



Suppose, radius of curvature of lens = R
 air film thickness = t (at distance $OQ = r$
 from point of contact O)

For bright ~~light~~ rings,

$$2\mu t \cos \theta = (2n+1) \frac{\lambda}{2}, \quad n = 0, 1, 2, 3, \dots$$

θ is small, $\cos \theta \approx 1$
 $\mu = 1$ (air)

$$\therefore 2t = (2n+1) \frac{\lambda}{2} \quad \text{--- (1)}$$

For dark rings,

$$2\mu t \cos \theta = n\lambda, \quad n=0, 1, 2, 3, \dots$$

$$\therefore 2t = n\lambda \quad \text{--- (2)}$$

In the figure,

$$EP \times HE = OE \times (2R - OE)$$

$$r \times r = t(2R - t)$$

$$r^2 = 2Rt$$

$$2R - t = 2R \\ (\text{approx.})$$

$$\text{or, } t = \frac{r^2}{2R}$$

Newton rings

Substituting value of t in eq. ① & ②,

Bright rings

$$\frac{2r^2}{2R} = (2n+1) \frac{\lambda}{2}$$

$$r^2 = \frac{(2n+1) \cdot \lambda \cdot R}{2}$$

$$r = \sqrt{\frac{(2n+1) \cdot \lambda \cdot R}{2}}$$

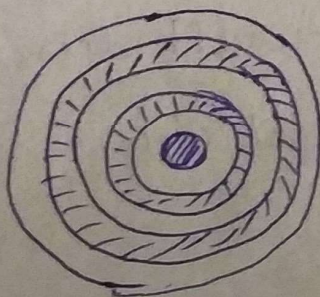
$$\boxed{r_n = \sqrt{\left(n + \frac{1}{2}\right) \lambda \cdot R}}$$

Dark rings

$$2 \times \frac{r^2}{2R} = n\lambda$$

$$r_n = \sqrt{nR\lambda}$$

When $n=0$, the radius of dark ring is zero and radius of bright ring is $\sqrt{\frac{\lambda R}{2}}$. Therefore the centre is dark and alternative bright/dark rings are produced.



While counting the order of dark rings 1, 2, 3 etc the central ring is not counted.

Therefore for the first dark ring, $n=1$

Diameter of n^{th} $D = 2 \cdot r$

$$D_1 = 2 \sqrt{\lambda R}$$

For $n=2$, $D_2 = 2 \sqrt{2 \lambda R}$

$$\therefore D_n = 2 \sqrt{n \lambda R}$$

Example

$$D_{16} = 2 \sqrt{16 \lambda R} = 8 \sqrt{\lambda R}$$

$$D_9 = 2 \sqrt{9 \lambda R} = 6 \sqrt{\lambda R}$$

Difference,

$$D_{16} - D_9 = (8 - 6) \sqrt{\lambda R}$$
$$= 2 \sqrt{\lambda R}$$

Similarly,

$$D_4 - D_1 = 2 \sqrt{4 \lambda R} - 2 \sqrt{\lambda R}$$
$$= 2 \sqrt{\lambda R}$$

Rings of higher order are observed closer to each other and finally seems to be merge.

Newton's rings by transmitted light

bright fringes,

$$2\mu t \cos \theta = n\lambda$$

$$\cos \theta = 1, \mu = 1$$

$$2t = n\lambda \quad \text{--- (1)}$$

For dark fringes

$$2\mu t \cos \theta = (2n+1)\frac{\lambda}{2}$$

$$\text{or } 2t = (2n+1)\frac{\lambda}{2} \quad \text{--- (2)}$$

$$\text{taking, } t = \frac{r^2}{2R}$$

from eq. (1),

$$r^2 = n\lambda R$$

$$r_n = \sqrt{n\lambda R}$$

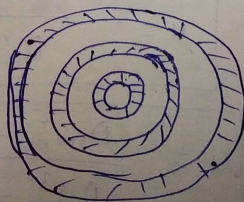
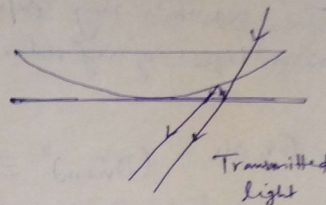
--- (3) (Bright fringe ring)

from eq. (2),

$$\frac{2 \times r^2}{2R} = (2n+1)\frac{\lambda}{2}$$

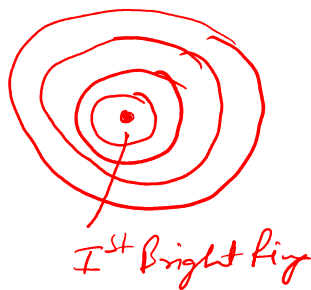
$$r = \frac{\sqrt{(2n+1)\lambda R}}{2} \quad \text{(dark ring)}$$

for $n=0$, $r_n = 0$ (Bright ring) $\rightarrow$ Centre is bright



Applications of Newton's Rings

- To test the flatness of a transparent glass surface.
- To find out wavelength of a monochromatic light.
- To find out refractive index of a liquid.



$$r_n = \sqrt{nR\lambda}$$

$$D_n = 2 \cdot r_n = 2\sqrt{nR\lambda}$$

$$D_n^2 = 4nR\lambda \quad \text{--- (1)}$$

$$D_1, D_2, D_3, D_4, \dots, D_n, D_{n+1}, \dots, D_{n+m}$$

$$D_{n+m}^2 = 4(n+m)R\lambda \quad \text{--- (2)}$$

$$D_{n+m}^2 - D_n^2 = 4mR\lambda$$

$$\lambda = \frac{D_{n+m}^2 - D_n^2}{4mR}$$

Refractive index of a liquid using Newton's rings.

The Newton's ring experimental arrangement is kept in a liquid of ref. index μ .

For air, $(D_{n+m})^2 = 4(n+m)\lambda R$ dark rings
 $D_n^2 = 4n\lambda R$
 $\therefore D_{n+m}^2 - D_n^2 = 4m\lambda R \quad \text{--- (1)}$

For liquid,

$$R^2 = \frac{n\lambda R}{\mu}$$
$$\therefore D^2 = \frac{4n\lambda R}{\mu}$$
$$D_{n+m}'^2 = \frac{4(n+m)\lambda R}{\mu}$$
$$D_n'^2 = \frac{4n\lambda R}{\mu}$$
$$D_{n+m}'^2 - D_n'^2 = \frac{4m\lambda R}{\mu} \quad \text{--- (2)}$$

$\therefore$ divide (1)/(2)

$$\mu = \frac{4m\lambda R}{D_{n+m}^2 - D_n^2}$$

$$\mu = \frac{D_{n+m}^2 - D_n^2}{D_{n+m}'^2 - D_n'^2}$$

Reading Advice:

Students, kindly read and prepare notes of reflective and anti-reflective coatings on glass surfaces following thin film interference. Subsequently, find out expression of coating thickness.